Exercise 4a

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Conduct task analysis for an app (e.g., online shopping) and document user flows. Create corresponding wireframes using Lucidchart.

AIM:

To understand and document the steps a user takes to complete the main tasks within an online shopping app.

Tool Link: https://www.lucidchart.com/pages/

PROCEDURE:

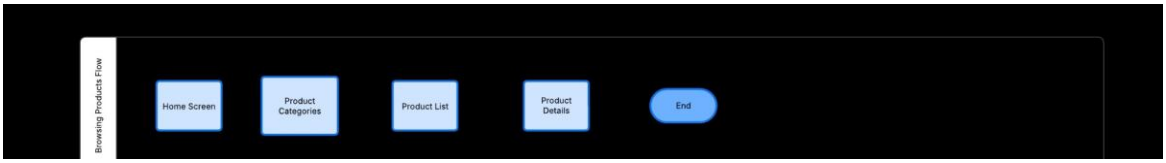
Step 1: Assigning Tasks

1. Browsing Products
2. Searching for a Specific Product
3. Adding a Product to the Cart
4. Checking Out

Step 2: Document User Flows

FRAME 1:

BROWSING PRODUCTS FLOW:



Analysis of the Browsing Products Flow

This flow represents how a user navigates the online shopping app to browse products without a specific item in mind. It begins at the Home Screen and ends at the Product Details page.

1. Purpose of the Screen

- This flow maps the visual exploration path a user takes from landing on the app to viewing a specific product.
- It tests how intuitively users can navigate category-based browsing without using the search function.

2. Key Elements and UI Components

- Home Screen –

    - User lands on the home page with featured products and category tiles prominently displayed.
    - Acts as the starting point for all user journeys within the app.
- Product Categories –
    - User taps on a category tile (e.g., Electronics, Clothing) to filter the product list.
    - Category labels and icons help users quickly identify the section they want.
- Product List –
    - User scrolls through a grid or list of products within the selected category.
    - Each card shows a product image, name, and price to aid quick decision-making.
- Product Details –
    - User taps on a specific product to view its full description, images, price, and reviews.
    - This is the final step before the user decides to add the item to the cart.
- End –
    - The oval End node marks the completion of the Browsing Products task flow.

3. How the Browsing Flow Works

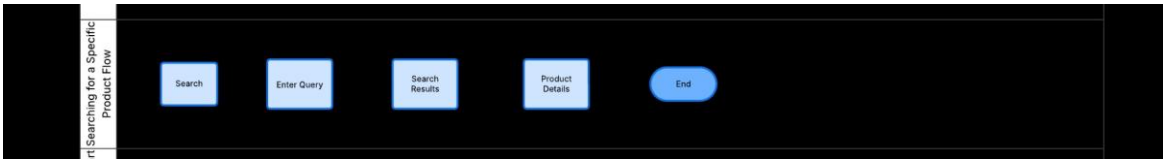
1. User opens the app and lands on the Home Screen.
2. User taps a product category to view relevant items.
3. User scrolls through the Product List to find items of interest.
4. User taps on a product to open the Product Details screen.

4. Cognitive and UX Benefits

- Category-based navigation reduces the number of items shown at once, preventing cognitive overload.
- The linear flow (Home → Categories → List → Details) matches users' natural mental model of browsing a store.
- Product cards with visual thumbnails allow users to scan faster than text-only lists.
- Each step progressively narrows the content, helping users focus without feeling overwhelmed.

FRAME 2:

SEARCHING FOR A SPECIFIC PRODUCT FLOW:



Analysis of the Search Flow

This flow captures the steps a user takes when they already know what product they are looking for. Instead of browsing categories, the user directly uses the search feature to locate the item quickly.

1. Purpose of the Screen
   - This flow documents the goal-directed search path, which is faster than category browsing for users with a specific product in mind.
   - It evaluates how efficiently users can locate products using keyword-based search.

2. Key Elements and UI Components
   - Search –
     - User taps the search bar or magnifying glass icon at the top of the screen.
     - This is a familiar interaction pattern found across all major shopping apps.
   - Enter Query –
     - User types a product name, brand, or keyword into the search input field.
     - Auto-suggestions may appear to reduce typing effort and guide users.
   - Search Results –
     - App displays a filtered list of products matching the entered query.
     - Users can apply filters (price, rating, brand) to narrow results further.
   - Product Details –
     - User taps on the desired product from the results to view full details.
     - Same Product Details screen as in the Browsing flow, ensuring UI consistency.
   - End –
     - The End node marks the completion of the Search flow.

3. How the Search Flow Works
   1. User taps the search bar or icon on any screen.
   2. User types a product name or keyword into the input field.
   3. App returns a list of matching products in the Search Results screen.
   4. User taps on a result to open the Product Details page.

4. Cognitive and UX Benefits
   - Direct search reduces the number of steps compared to category browsing, improving efficiency for goal-directed users.
   - Auto-suggestions lower the cognitive effort of remembering exact product names.
   - Displaying search results in a familiar list format (same as Product List) maintains interface consistency.
   - The flow mirrors real-world shopping behaviour, making it immediately learnable for new users.

FRAME 3:
ADDING A PRODUCT TO THE CART FLOW:

Analysis of the Add to Cart Flow

This flow documents the steps a user takes to select a product and add it to their shopping cart. It extends the browsing or search flow by including the product configuration step before the cart action.

1. Purpose of the Screen
   - This flow captures the product selection and cart addition process, which is the core transactional step in any shopping app.
   - It evaluates whether users can successfully configure their choices (size, colour, quantity) before adding an item.

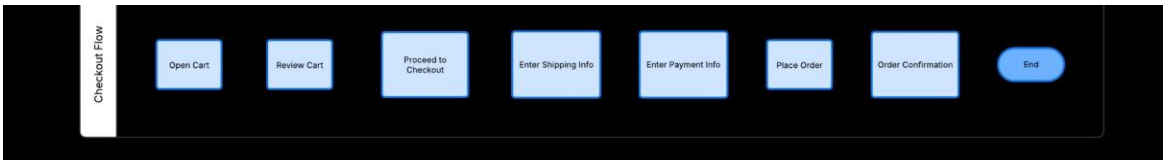
2. Key Elements and UI Components
   - View Products –
     - User arrives at a product through browsing categories or using the search bar.
     - This step represents the entry point shared with both earlier flows.
   - Product Details –
     - User taps on the product to open the full details page with images, description, and price.
     - The Add to Cart button is prominently placed on this screen.
   - Select Options (size, colour, quantity) –
     - User selects the specific variant they want before adding the item to the cart.
     - Drop-downs or swatches are used for colour and size; a stepper control manages quantity.
   - Add to Cart –
     - User taps the “Add to Cart” button to place the selected item in the cart.
     - A confirmation toast or badge update provides immediate feedback.
   - End –
     - The End node marks the successful completion of the Add to Cart task.

3. How the Add to Cart Flow Works
   1. User browses or searches and taps on a product to open its details.
   2. User selects the desired size, colour, and quantity from the available options.
   3. User taps “Add to Cart” to add the configured product to the shopping cart.
   4. App confirms the action and updates the cart icon badge count.

4. Cognitive and UX Benefits

- Grouping product options (size, colour, quantity) on a single screen reduces the number of steps before adding to cart.
- Immediate visual feedback (badge update, toast notification) confirms the action without requiring the user to navigate away.
- The “Add to Cart” button is consistently positioned and styled, making it easy to locate across different product pages.
- Keeping the selection step before the cart action prevents unwanted additions and reduces the need for cart editing.

FRAME 4:
CHECKOUT FLOW:



Analysis of the Checkout Flow
This flow represents the most complex task in the online shopping app. It takes the user from reviewing the cart through to placing the order and receiving a confirmation, covering payment and shipping information entry.

1. Purpose of the Screen
   - This flow documents the complete order placement journey, which is the most critical and error-sensitive path in any e-commerce application.
   - It evaluates how well the app guides users through multiple sequential data-entry steps without causing drop-off.

2. Key Elements and UI Components
   - Open Cart –
     - User taps the cart icon to review all selected items before proceeding to checkout.
     - The cart screen summarises product names, quantities, prices, and the total amount.
   - Review Cart –
     - User verifies the items in the cart and can edit quantities or remove products.
     - A clear “Proceed to Checkout” button guides the user to the next step.
   - Proceed to Checkout –
     - User taps “Checkout” to begin the order placement process.
     - The app may prompt login or guest checkout at this stage if the user is not signed in.
   - Enter Shipping Info –
     - User provides their delivery address, including street, city, state, and postal code.
     - Saved addresses can be retrieved to reduce manual entry for returning users.
   - Enter Payment Info –
     - User enters card details or selects a saved payment method (UPI, wallet, card).
     - Security indicators (lock icons, SSL labels) reassure users during this sensitive step.

- Place Order –
  - User reviews the final order summary and taps “Place Order” to confirm the purchase.
  - This is the point of no return — a confirmation dialog may appear to prevent accidental taps.
- Order Confirmation –
  - App displays the order ID, estimated delivery date, and a success message.
  - User may receive an email or SMS confirmation simultaneously.
- End –
  - The End node marks the successful completion of the entire checkout process.

3. How the Checkout Flow Works

5. User opens the cart and reviews all selected products.
6. User taps “Proceed to Checkout” to begin the order placement process.
7. User enters or confirms their shipping address.
8. User enters payment details or selects a saved payment method.
9. User reviews the final order summary and taps “Place Order”.
10. App displays the Order Confirmation screen with order ID and delivery estimate.

4. Cognitive and UX Benefits

- Breaking the checkout into sequential steps (shipping → payment → review) reduces cognitive load by presenting one decision at a time.
- Progress indicators at the top of each checkout screen help users understand how many steps remain, reducing anxiety.
- Saving shipping and payment details for returning users dramatically shortens the checkout flow on repeat visits.
- The Order Confirmation screen provides closure, reassuring users that their purchase was successful and reducing the urge to re-submit.

LUCIDCHART LINK:
https://www.lucidchart.com/pages/